

Designing a Virtual Death VR Experience

Stefan Greuter

Deakin Motion Lab Deakin University
School of Communication and Creative Arts
Geelong, Victoria, Australia
stefan.greuter@deakin.edu.au

Gerard T. Mulvany

Deakin Motion Lab Deakin University
School of Communication and Creative Arts
Geelong, Victoria, Australia
g.mulvany@deakin.edu.au



Figure 1: Installation view of Shaun Gladwell's *Passing Electrical Storms* 2023 on display as part of the Melbourne Now exhibition at The Ian Potter Centre: NGV Australia, Melbourne from 24 March – 20 August 2023. Image: Sean Fennessy

Abstract

Virtual Reality (VR) exhibitions are becoming increasingly popular for creating immersive and interactive experiences. One particular use case in research is death simulations in VR, where users can experience or explore the concept of death in a controlled environment. Death is a profound theme that offers opportunities for emotional growth, empathy, and deep reflection. This project presents *Passing Electrical Storms*, an innovative virtual reality art installation by Shaun Gladwell, designed to evoke the human experience of dying and an out-of-body journey through the universe. The installation combines VR, sensing technologies, and tactile feedback to create a visually stunning and emotionally immersive experience. Participants journey from a simulated cardiac arrest to brain death, moving through inner-body experiences and vast cosmic expanses, provoking contemplation on mortality and the sublime nature of the universe. This work demonstrates the potential of VR to engage users with complex emotional and existential themes, providing valuable insights for future VR design in the field of HCI.

CCS Concepts

• **Human-centered computing** → **User centered design.**

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Virtual Reality, Simulated Death, Mortality, Immersive Experience, Sensory Feedback, Haptic Feedback, Out-of-Body Experience, Artistic Visualization, Emotional Immersion, Interactive Art

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1 Introduction

Death is a crucial concept for humanity, as it represents an inescapable aspect of the human experience with profound implications for our understanding of life, emotions, and relationships. The idea of death and what happens after death is primarily shaped by cultural and spiritual beliefs and has a significant impact on personal growth and existentiality. Recognizing the inevitability of death can sometimes lead to a greater appreciation for life, potentially fostering introspection, resilience, and motivation to make the most of our limited time [2].

Exploring the concept of death in VR offers users a unique, immersive experience that may enable them to safely engage with mortality-related scenarios, fostering empathy and self-reflection. By simulating end-of-life situations or exploring themes of loss and grief [5], VR applications might promote self-reflection and encourage users to confront their own feelings and fears about death [6]. These immersive experiences could yield several psychological benefits, including reduced death anxiety and an increased sense of spiritual well-being [4]. By providing a controlled environment for exposure to mortality-related themes, VR applications

can help users gradually acclimate to and process these difficult emotions [16]. Moreover, VR technology offers innovative educational resources for users to better understand end-of-life care [8] and advanced care planning [17]. It also considers the emotional and psychological impact, ethical and legal concerns regarding the digital resurrection of deceased individuals [5], and issues related to integrating technology into end-of-life care and simulations [15]. By facilitating firsthand experiences of different aspects of death and dying, VR can promote a deeper and more nuanced understanding of these topics, ultimately leading to more informed and empathetic perspectives [13]. Users might benefit from the psychological advantages of reduced death anxiety and enhanced overall well-being while also accessing innovative educational resources [8]. Furthermore, VR could provide valuable support for end-of-life care and therapy, allowing users to better navigate and understand their own or their loved ones' experiences related to mortality and end-of-life decisions [15].

Artworks have started to employ VR to explore themes of mortality, grief, and the impermanence of life, offering unique and thought-provoking experiences for users. The artworks that informed our design employed VR technology as a means to address the concept of mortality and themes of grief [14] and suicide [18].

Passing Electrical Storms investigated how art, design, and technology can facilitate a dialogue around mortality and the fleeting, precious nature of life itself. It ruminated on the notion of the scientific sublime by imagining the process of individual death within a universal frame with no known limit or end. The project investigated immersive technology, sensors, and tactile technology to develop an experiential pathway from life to death, meditating on the universes within and beyond the body.

2 Implementation

The installation aimed to capture audiences with an interactive, aesthetically pleasing, and highly functional sensorial immersive journey. The exhibition was designed to allow everyone to access and immerse themselves in the art installation, regardless of their unique physical requirements. One station was reserved to cater to visitors who use mobility aids. The design addressed functionality and safety within the exhibition environment by integrating easily cleanable surfaces on VR headsets, beds, and UVC cleaners at each station, as illustrated in Figure 1. Furthermore, controls were embedded into each station to stop the simulation and adjust the experience's haptic intensity.

Following design guidelines for public VR experiences [11], the installation space was created with a holistic user journey in mind. The VR setup was in a designated space within the gallery, with the environment designed to resemble a hospital room equipped with heart rate monitors and VR headsets. Visitors were first greeted with an atmosphere reminiscent of a hospital space, priming them for the coming experience. Clear explanatory text outlined the process, while well-trained invigilators facilitated visitor screening and participant engagement with the work, as suggested by prior research [8].

The experience aimed to provide a meditative and introspective simulation of death that would provoke visitors to contemplate their mortality and the essence of life, as suggested by Bahng et al. [1].

This orchestrated arrangement of sensory stimuli was intended to deliver a truly immersive out-of-body journey. After the experience, invigilators supported visitors in returning safely to the exhibition space, ensuring a holistic and smooth journey from start to finish.

2.1 Design Approach

During concept development, the team grappled with the delicate balance between crafting an immersive death experience using fear-arousal [12] without causing excessive distress to visitors. Research on palliative care [8] informed our responsible approach and influenced the interaction design of the work, leading to the inclusion of safety mechanisms to reduce the intensity of the haptic experience and controls for invigilators to stop the simulation and allow visitors to exit at any point. We also conceptualized the VR environment by experimenting with multiple sensory elements such as visual aesthetics, haptic feedback, and sound design to create an immersive and transformative experience.

We involved participants in the user-testing phase to ensure safety and comfort, making adjustments based on their feedback through several iterations. This was pivotal in creating an experience that was not only technically sound but also emotionally resonant. This process allowed us to gauge user responses to the VR experience and, importantly, identify and mitigate any possible triggers of anxiety or panic.

2.2 Technology Use and Implementation

The VR experience combined custom-built and off-the-shelf technologies to create a cohesive system that seamlessly blended physical sensations with virtual experiences, emphasizing aesthetic and emotional impact. The following section explores the game engine configuration, heartbeat sensor integration, haptic feedback mechanisms, the use of moving air to cool down the participant and provide a sense of moving through space, the implementation of a self-reflecting camera, and our commitment to environmentally sustainable design. Each component was designed to enhance the installation's immersive quality and artistic vision, fulfilling Gladwell's goal of creating a visually and emotionally profound experience.

2.2.1 System Architecture. The game engine was configured to receive serial input communication from Arduino or other peripheral devices. This serial I/O stream was decoded by the game engine to interpret commands and data, sending responses back through the same stream. This communication was facilitated by a USB 1.0 connection to an Arduino Uno board. The Arduino board acted as the central distributor of commands from the I/O stream, effectively making the game engine the commanding voice of the system. The Arduino board sent commands to several peripheral devices, enabling specific sensory experiences and facilitating bodily sensing through heart rate sensors. The Arduino board also mediated between physical controls and the game engine, allowing peripheral input devices to control the game engine alongside the standard mouse and keyboard setup. This configuration provided invigilators with multiple methods for starting, resetting, and aborting the experience, whether through the tactile buttons mounted on the side of the bed or keyboard commands.

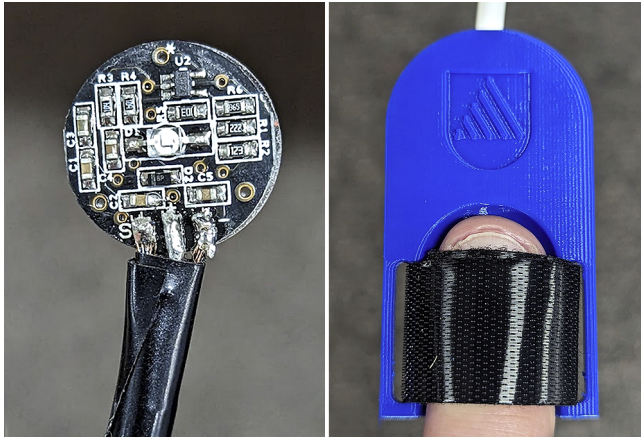


Figure 2: Heart rate sensor (left) embedded in 3D printed finger clamp (right)

2.2.2 Heartbeat Sensor Integration. Critical to Gladwell’s vision was the synergy between the real and virtual worlds, creating an immersive environment where participants felt embedded in the virtual reality. This intersection was intended to allow participants to fully engage with the artwork and its visualization, fostering emotional vulnerability. The heartbeat sensor was a pivotal component, originally intended to drive the opening cinematic, where participants experienced a heart attack followed by their virtual death and the beginning of the out-of-body experience. The real-time display of the participant’s heartbeat on-screen created a link to the physical world, enhancing cognitive dissonance during the simulated heart attack. We used an infrared heartbeat sensor (Figure 2, left) designed for Arduino (<https://pulsesensor.com/>), which returned luminance values and beats per minute (BPM) data. This data was visually represented as an ECG, contributing to the immersive experience.

Due to the noisy nature of the raw data, we implemented conditions within the Arduino code to filter out noise and ensure consistent heart-rate detection. If the BPM was unrealistic, the system would switch to pre-recorded data to maintain the visual integrity of the experience, preventing distractions for the user. Additionally, we modified the heart-rate sensor for durability, replacing its original wiring with XLR cables for their toughness and easy availability. The sensor was enclosed in a custom-designed 3D-printed case. Velcro straps secured the sensor to the participant’s finger (Figure 2, right). While wireless heart-rate sensors are available, we chose a wired version to prevent loss, ensure cost-effectiveness, and allow for easy replacement during the exhibition.

2.2.3 Enhancing Haptic Engagement. To further enhance immersion, the heartbeat data was used to synchronize haptic feedback with the participant’s real heartbeat. A 100-watt haptic transducer was mounted underneath the chest section of the bed, connected to a miniature amplifier with adjustable volume controls illustrated in Figure 3. This setup allowed the participant or gallery staff to modify the haptic intensity. The amplifier received audio input from an MP3 chip that played a heartbeat audio file stored on a microSD card. The Arduino triggered this audio playback in sync



Figure 3: Arduino-powered control panel for pausing, restarting, and adjusting haptic intensity

with the detected heartbeat. At a specific point in the virtual reality experience, the game engine instructed the Arduino to switch from real-time heartbeat data to pre-recorded heart attack data, aligning the haptic feedback with the visual simulation of a heart attack in the virtual world.

Similarly, Gladwell envisioned modulating both the skin sensations and temperature sensations of the participant. He envisaged the experience beginning with the participant feeling progressively warmer as the heart attack simulation commenced. Fans would then activate, gradually cooling the body as the participant’s spirit or body appeared to float through the universe. Initially, electronically controlled heaters that provided heat at the start and then were turned off at the point of virtual death were considered to enhance temperature modulation. However, this idea was deemed infeasible due to cost and potential safety risks, and the focus shifted solely to using fans.

The haptic bed illustrated in Figure 4 on which participants lay featured numerous perforations resembling an air hockey table, but with larger holes. The bed was designed and laser-cut with an internal skeletal structure to facilitate airflow and accommodate 200-millimeter computer fans. These PWM (Pulse Width Modulation) fans allowed speed control via the Arduino. The final design incorporated six 200-millimeter RGB PWM computer fans connected to a fan controller, which, in turn, was linked to the Arduino. The Arduino received RPM (Revolutions Per Minute) control commands from the game engine, similar to the other devices discussed earlier. This design allowed high-pressure air accumulated in the underside cavity of the bed to rush out against the participant’s skin, creating sensations of airflow and cooling the skin temperature, as Gladwell had envisioned.

2.2.4 Self-Reflection in the Immersive Environment. One of the most critical elements in creating the immersive connection between the participants’ physical sensations and the virtual experience was the creation of a self-reflection of the user lying on the bed using a camera mounted above. This feature aimed to make participants feel as if they were floating above their bodies, watching themselves from a few meters away. A Logitech Brio webcam positioned above the bed was connected via USB 3.0 extension cables to the computer

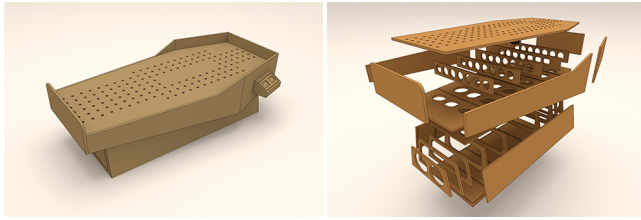


Figure 4: Haptic Bed design (left) and exploded view (right) designed by Leigh Russell featuring wind effects powered by fans and heart rate vibrations via a transducer

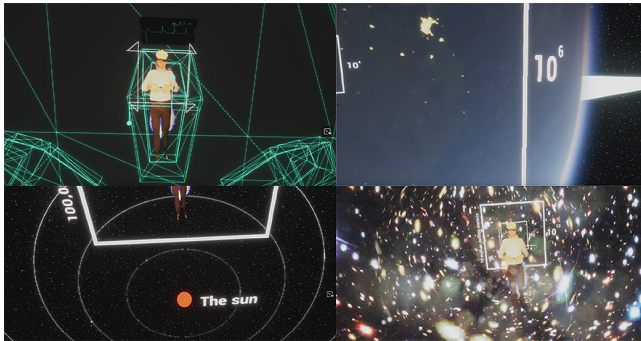


Figure 5: User journey from the exhibition environment to the outer edges of the universe

running the game engine and the virtual reality experience. The open-source software OBS (Open Broadcaster Software) processed the webcam input, adjusted the video dimensions, and applied a masking effect using a solid blue background. This video feed was then sent to the game engine, which used chroma keying to remove the blue background, necessitating the room and beds to be designed in a deep, rich blue. The video, now with an alpha mask, was imported into Unreal Engine’s material system, where the luminance of the image created a tessellated displacement effect, adding subtle depth and three-dimensionality to the participant’s body image. Gladwell considered this element crucial to the overall experience, noting that tests without the video were far less effective in achieving his artistic goals.

2.2.5 VR Experience. Gladwell drew inspiration from the famous short film *Powers of 10* by Charles and Ray Eames and conducted extensive research into philosophical and scientific perspectives on death, drawing from thinkers like Baudrillard [3], Foucault [9, 10], and Chalmers [7]. This research informed the conceptual design of the experience and exhibition space, detailed in a briefing document. In the VR experience illustrated in Figure 5, users embark on their end-of-life journey through a scientific visualization of the universe. Starting in the exhibition environment, participants float at exponentially increasing speeds, visualizing the known universe. This journey provides a space for participants to contemplate their mortality in a deeply personal and evocative manner, rendered as a necro-nautical self-portrait.

2.2.6 Environmentally Conscious Design. The design embraced circular economy principles, prioritizing reuse and minimizing waste. We repurposed discarded wiring from old computers, audio systems, and televisions, saving over 20 meters of wiring. The 3D-printed cases were made from compostable locally sourced, recycled sugarcane PLA filament. Using Arduino as the serial I/O hub, we ensured the bed hardware was connected to the VR system via a single USB, enhancing versatility and allowing easy software updates.

3 Discussion

The design of *Passing Electrical Storms* explores the themes of mortality and the human experience from a novel perspective through a deeply immersive and introspective VR experience. By incorporating detailed sensory feedback mechanisms and safety features, the project balances immersive impact and user comfort. Notably, the inclusion of heartbeat sensors, haptic feedback, and temperature modulation through wind effects aimed to stimulate multiple senses, thereby increasing the emotional depth and connection with the VR experience. This provided users with a profound journey from life to death. However, some aspects, such as the initial implementation of temperature modulation through electronically controlled heaters, proved infeasible due to cost and safety concerns, demonstrating the need for practical adjustments in the design process. One surprising and intriguing outcome was the effectiveness of the self-reflection feature, which created a sense of out-of-body experience for users and was crucial in achieving the project’s artistic goals. Conversely, it was equally surprising that the impact of the wind effect generated by the fans in the bed was not as noticeable to most users. While critics, the NGV, and Shaun Gladwell, as well as preliminary interview data from users, suggest that the prototype successfully engaged users in a contemplative exploration of mortality, future research should continue to refine these design elements to enhance the overall experience, helping to offer innovative and impactful VR experiences that resonate on a deeply personal level.

4 Conclusion

Passing Electrical Storms advances VR to explore profound human experiences such as mortality, offering users a visually stunning and emotionally immersive journey that encourages deep reflection and introspection. This project demonstrates VR’s potential to create contemplative spaces by integrating artistic visualization, haptic feedback, and sensory modulation. The user-centered design in creating aesthetically impactful VR experiences is underpinned by wearable technology including a VR headset and heartbeat sensor and vibrations that can be felt throughout the user’s body. As we continue to innovate, VR will offer new opportunities in education, therapy, and beyond, enhancing our understanding of the human condition through engagement with profound existential themes.

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References

- [1] Sojung Bahng, Ryan M. Kelly, and Jon McCormack. 2020. Reflexive VR Storytelling Design Beyond Immersion: Facilitating Self-Reflection on Death and Loneliness. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (New York, NY, USA). ACM, 1–13. <https://doi.org/10.1145/3313831.3376582>
- [2] Itxaso Barberia, Ramon Oliva, Pierre Bourdin, and Mel Slater. 2018. Virtual mortality and near-death experience after a prolonged exposure in a shared virtual reality may lead to positive life-attitude changes. *PLOS ONE* 13 (11 2018), e0203358. Issue 11. <https://doi.org/10.1371/journal.pone.0203358>
- [3] Jean Baudrillard. 2019. Simulacra and simulations (1981). In *Crime and Media*. Routledge, 69–85.
- [4] Mackenzie Blomstrom, Andrew Burns, Daniel Larriviere, and Jennifer Kim Penberthy. 2022. Addressing fear of death and dying: traditional and innovative interventions. *Mortality* 27 (1 2022), 18–37. Issue 1. <https://doi.org/10.1080/13576275.2020.1810649>
- [5] Amber Boothe. 2022. The death and life of Jang Nayeon: a case for personality rights in the digital layers of reality. *International Journal of Law and Information Technology* 30 (12 2022), 398–422. Issue 4. <https://doi.org/10.1093/ijlit/eaad005>
- [6] Pierre Bourdin, Itxaso Barberia, Ramon Oliva, and Mel Slater. 2017. A Virtual Out-of-Body Experience Reduces Fear of Death. *PLOS ONE* 12 (1 2017), e0169343. Issue 1. <https://doi.org/10.1371/journal.pone.0169343>
- [7] David J Chalmers. 1997. *The conscious mind: In search of a fundamental theory*. Oxford Paperbacks.
- [8] Daniel Eckhoff, Royce Ng, and Alvaro Cassinelli. 2022. Virtual Reality Therapy for the Psychological Well-being of Palliative Care Patients in Hong Kong. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 1–5. <https://doi.org/10.1109/ISMAR-Adjunct57072.2022.00010>
- [9] Michel Foucault. 1965. Madness and Civilization: A History of Insanity in the Age of Reason, trans. Richard Howard (New York: Pantheon, 1965) 16 (1965).
- [10] Michel Foucault. 2002. *The birth of the clinic*. Routledge.
- [11] Stefan Greuter, Florian 'Floyd' Mueller, and Thuong Hoang. 2022. Designing public VR installations. In *Designing Interactive Systems Conference* (New York, NY, USA). ACM, 792–806. <https://doi.org/10.1145/3532106.3533454>
- [12] Thuong Hoang, Stefan Greuter, Simeon Taylor, George Aranda, and Gerard T Mulvany. 2021. An evaluation of virtual reality for fear arousal safety training in the construction industry. In *2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 177–182.
- [13] Victor Kaptelinin. 2016. Making the case for an existential perspective in HCI research on mortality and death. In *Conference on Human Factors in Computing Systems - Proceedings*, Vol. 07-12-May-2016. <https://doi.org/10.1145/2851581.2892585>
- [14] Signal Space Lab. 2019. Afterlife [Virtual reality game]. <https://www.afterlife-vr.com/>
- [15] Michael Massimi and Andrea Charise. 2009. Dying, death, and mortality. In *CHI '09 Extended Abstracts on Human Factors in Computing Systems* (New York, NY, USA). ACM, 2459–2468. <https://doi.org/10.1145/1520340.1520349>
- [16] Silvia Francesca Maria Pizzoli, Dario Monzani, Laura Vergani, Virginia Sanchini, and Ketti Mazzocco. 2023. From virtual to real healing: a critical overview of the therapeutic use of virtual reality to cope with mourning. *Current Psychology* 42 (4 2023), 8697–8704. Issue 11. <https://doi.org/10.1007/s12144-021-02158-9>
- [17] Madison B. Smith, Tamara G. R. Macieira, Michael D. Bumbach, Susan J. Garbutt, Sandra W. Citty, Anita Stephen, Margaret Ansell, Toni L. Glover, and Gail Keenan. 2018. The Use of Simulation to Teach Nursing Students and Clinicians Palliative Care and End-of-Life Communication: A Systematic Review. *American Journal of Hospice and Palliative Medicine* 35 (8 2018), 1140–1154. Issue 8. <https://doi.org/10.1177/1049909118761386>
- [18] Paisley Smith. 2019. Homestay. <https://www.paisleymsmith.com/homestay>